

Stackable Citation Knowledge

Building on Nanopublications for Climate and Biodiversity Research

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A collaboration between VitenHub AS · Knowledge Pixels · Prophet Town

Citation data has two jobs

Credit prior work · works today · Enable building upon · open problem

A reference list entry today

Soroye, P., Newbold, T., & Kerr, J. (2020).
Climate change contributes to widespread
declines among bumble bees across continents.
Science, 367, 685.
doi:10.1126/science.aax8591

→ doi.org/10.1126/science.aax8591

The reader still has to read the body text to learn *which* of Soroye's claims this paper builds on, whether that claim was confirmed, disputed, or qualified, and under which conditions.

A stackable reference list entry

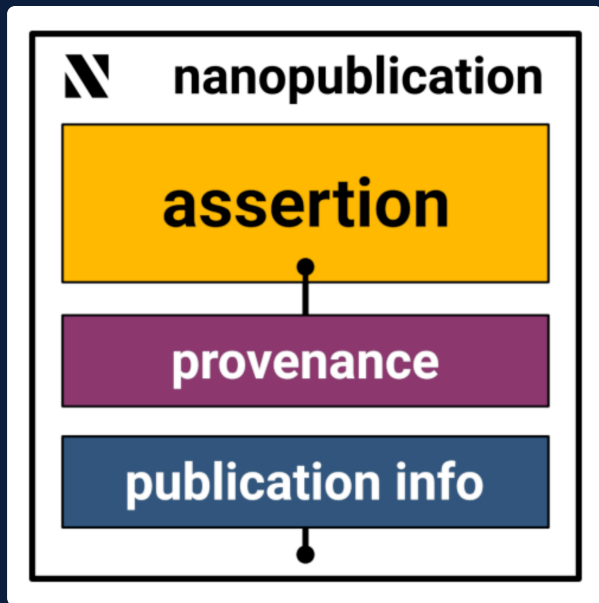
The screenshot shows a web interface for a citation. At the top, there are tabs for 'Template View' and 'RDF View'. Below the tabs, the citation is displayed in a structured format. The citation text is: 'The Creative or Scholarly work'. Below this, there is a line of text: 'TEI-based extirpation projection is grid-coupled at the projection step (Iberian Bombus, HEALPix nsidc=64 vs nsidc=128, SSP3-7.0): mechanism replicates at fit time, per-species ranking does not — fixable'. Below this, there is a section titled 'cites the following:'. This section contains three entries, each with a relationship type and a URL. The first entry is 'extends' with the URL 'https://doi.org/10.5281/zenodo.20113777'. The second entry is 'extends' with the URL 'https://doi.org/10.5281/zenodo.20113780'. The third entry is 'qualifies' with the URL 'Climate change contributes to widespread declines among bumble bees across continents'.

→ [CiTO Nanopub](#) · [Research Synthesis Nanopub](#)

The new study replicate the claim from the original paper, records each step via nanopublications and produces the qualification of the cited paper.

OpenCitations + WikiCite already build the left. We add a citable target on the right.

Stackable architecture



Why stackable

- Each step in a citation chain is **its own nanopub**
- Each nanopub has **its own URI**
- A future paper can cite **one specific claim**, not just the paper
- CiTO types make the *intent* machine-readable:

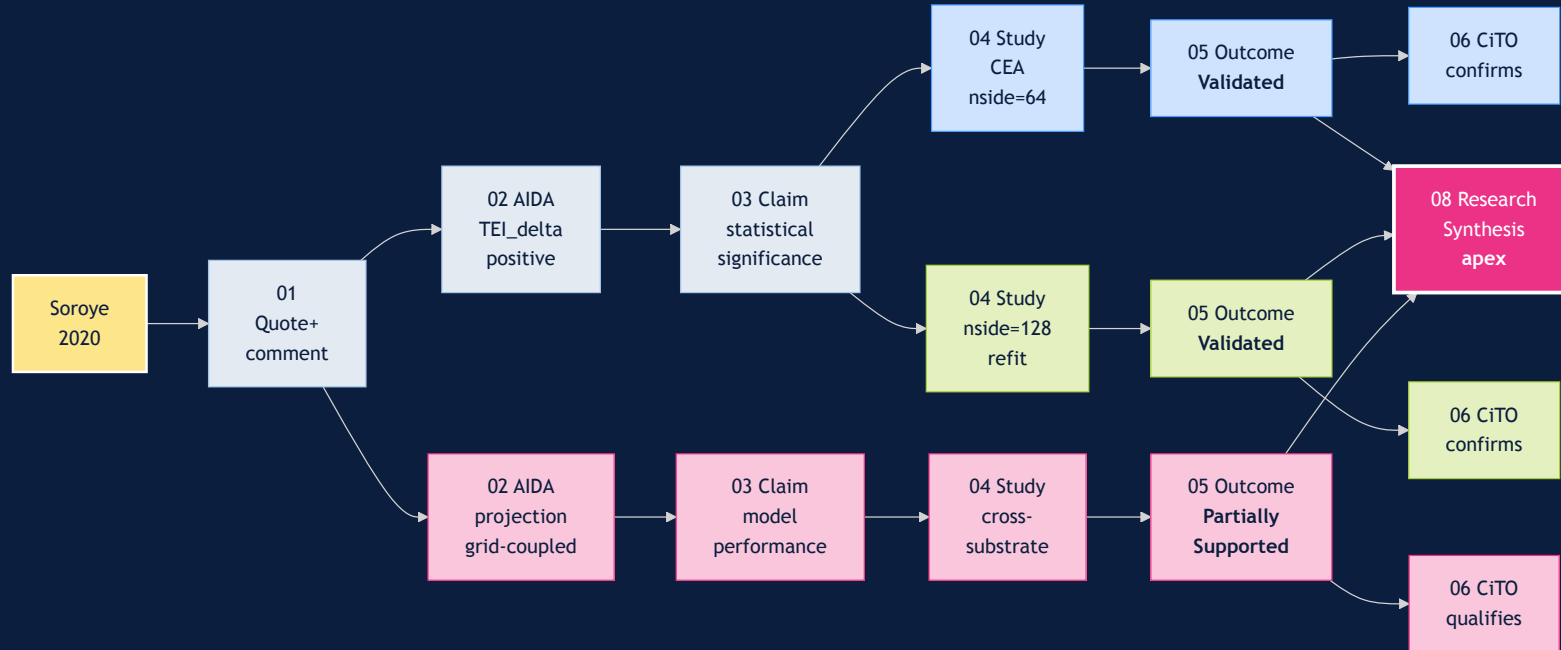
`extends · usesMethodIn · disputes ·
citesAsEvidence · qualifies`

Assertion · RDF triples + Wikidata Q-IDs **Provenance** · PROV-O derivation chain
Publication info · ORCID · timestamp · CC-BY 4.0

Stackable in practice — a FORRT replication chain

Soroye et al. 2020 (Science) · Iberian *Bombus* under SSP3-7.0 · apex [RA5TJVZ0...]

(https://w3id.org/sciencelive/np/RA5TJVZ0_5Knxzd4OtOoZgO6ZspWHwVCSLWNNd7V9H6QQ) wraps the whole chain into one citable URI



All 16 nodes are addressable nanopubs · published on the open Science Live network.

Three production paths to a signed nanopub

Self-driving cars will drive themselves — but humans still choose the destination. Same for LLM citation extraction: the model does the work · the author chooses the intent.

Author

Prospective · author of citing work

Workflow: open paper in Zotero · pick template · pick CiTO relation · plugin signs + publishes.

Worked example: Soroye 2020 → Bombus chain · Quote-with-comment [RAErLL...](#)

Intent at *write time* — no ceiling.

Annotate

Mid-reading · third-party reader

Any expert reader declares the inferred CiTO relation between a paper they're reading and one of *its* references. Signed by the reader's ORCID, contestable, refinable.

Demoed next with Wernberg 2016 and one of its cited works.

Extract, Replicate, and Extend

Retroactive · semi-automated

Any expert declares a Research Question, use the PRISMA framework with LLM to select publications, declare the intended usage with CiTO, and replicate.

PDF → refs + context via PICO / PCC
→ FORRT Replication with LLM → C
→ signed nanopubs

Three paths · same destination · same nanopub network.

Three worked domains

One named citable URI per column — these resolve today on the open network

Biodiversity

LifeWatch ERIC

Iberian *Bombus* replication of Soroye
2020 — three branches under SSP3-7.0.

Apex Synthesis [RA5TJVZ0...](#)

Breadth: a PRISMA-compliant
systematic review on quantum-
biodiversity adds **238 more nanopubs**
with full CiTO typing.

Climate change

EarthCARE → DGGS pipeline

Full software stack converting
EarthCARE EO data to HEALPix-geo,
published as a FORRT chain + Research
Software nanopub.

CiTO citation [RAN0HJytFV4...](#)

Cross-discipline

FIESTA — plankton classification

Scattering transforms from
astrophysics (Delouis et al. 2022)
applied to **plankton imagery** (Decrop et
al. 2025) — +8 pp rare-class recall.

One CiTO nanopub, three predicates:

`extends` · `usesMethodIn` ·
`citesAsDataSource`

[RA8BSfq4Cbs3...](#)

On the network today: 18-nanopub *Bombus* chain · 238-nanopub PRISMA review · EarthCARE → DGGS chain · FIESTA cross-disciplinary chain.

Demo · annotate while reading

The screenshot displays the Zotero application window. At the bottom, a video player shows a progress bar at 0:00 / 5:05. The main interface is divided into three panels. The left panel, titled 'My Library', lists various collections, with 'MarineSystems' highlighted in blue. The middle panel shows a table of citations with columns for 'Title' and 'Creator'. The right panel displays the details for the selected citation, 'Climate-driven regime shift of a temperate marine ecosystem' by Wernberg et al.

Title	Creator
Climate-driven regime shift of a temperate marine ecosystem	Wernberg et al.

Climate-driven regime shift of a temperate marine ecosystem

Info

Item Type	Journal Article
Title	Climate-driven regime shift of a temperate marine ecosystem
Author	Wernberg, Thomas
Author	Bennett, Scott
Author	Babcock, Russell C.
Author	De Bettignies, Thibaut
Author	Cure, Katherine
	17 more...
Publication	Science
Publisher	
Place	
Date	2016-07-08
Volume	353
Issue	6295
Section	
Part Number	
Part Title	
Pages	169-172
Series	
Series Title	
Series Text	
Journal Abbr	Science
DOI	10.1126/science.aad8745
Citation Key	
URL	https://www.science.org/doi/10.1126/science.aad8745
Accessed	26/05/2026, 23:18:18
PMID	
PMCID	

Atomicity enables claim-level verification

*Hallucinated and miscited references are increasingly slipping through peer review — even in high-ranked journals. Reviewers are unpaid researchers, and many now use LLMs themselves. "Verify the whole paper" is intractable. **Verify one claim at a time.***

LLM-extracted

lowest

- Pipeline output from intent classifier
- Open to **intent-hallucination**
- *Example:* LLM-classified citation in the PRISMA review

Reader-annotated

mid-low

- Expert reader infers CiTO for *someone else's* citation
- Signed by reader's ORCID, contestable
- *Example:* Wernberg → Poloczanska annotation (demo)

Author-validated

mid-high

- Author of citing work declares intent at write time
- Signed nanopub binds intent to assertion
- *Example:* RAErLL... (slide 5)

FORRT-replicated

highest

- Replication links Outcome to Claim
- Trust becomes a **citation signal**
- *Example:* RAa4QR41... (slide 4)

*Do one thing and do it well — applied to citations. Atomicity makes verification **tractable**, not zero.*

Try it · Cite it · Extend it

Open source · open governance. Your nanopubs live in the decentralised network — not on our servers. Even if Science Live disappears, what you published stays addressable, forever.

Platform



platform.sciencelive4all.org

Open · CC-BY 4.0 · no login

Zotero plugin



ScienceLiveHub/.../zotero

Zotero 7+ · install + walkthrough

Replication scaffold



[ScienceLiveHub/forrt-replication-
template](https://ScienceLiveHub/forrt-replication-template)

Reproducibility · `DOMAIN.md`
adapts

This deck



[ScienceLiveHub/citex2026-
stackable-citations](https://ScienceLiveHub/citex2026-stackable-citations)

Slidev · demo · Zenodo DOI

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